Shubham Haraniya

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Education

- **M.Tech in Artificial Intelligence** – Indian Institute of Technology, Jodhpur *2025 – Present, CGPA: 9.0*
- **B.E. in Computer Engineering** – L.D. College of Engineering, Ahmedabad *2021 – 2025, CGPA: 8.49*

Experience

- **IntelKit Solutions** *Jan 2025 – June 2025*

Data Scientist Intern

  - Developed a **deepfake detection system** using computer vision and deep learning techniques to identify manipulated images and videos, implementing preprocessing, feature extraction, model training, and evaluation pipelines for reliable media authenticity analysis.
  - Designed and deployed automated data acquisition pipelines using Selenium and BeautifulSoup, enabling extraction and reducing manual data collection effort by **over 70%** for downstream NLP and Machine Learning workflows.

Projects

- **Deepfake Forensics – Frequency-Aware SE-ADDNet** *Github*

Deep Learning / Computer Vision Project Feb 2026 – April 2026

  - Proposed a novel **dual-stream architecture** jointly modeling spatial manipulation cues (EfficientNet-B5 + SE/ADD attention) and FFT-based spectral artifacts via a **learnable FrequencyGate fusion**, achieving **94.6% accuracy** and **96.3% F1** on 12K test images across 4 deepfake generators.
  - Served real-time **GradCAM forgery localization** via a Streamlit web app, providing interpretable region-level detection explanations.
  - **Technologies**: PyTorch, EfficientNet-B5 (timm), FocalLoss, FP16 Mixed Precision, GradCAM, Streamlit

- **Crop Disease Advisor – Vision + LLM Pipeline** *Github Demo*

MLOps / Applied AI Project Jan 2026 – Feb 2026

  - Built an end-to-end **vision-language pipeline** for plant disease diagnosis – EfficientNet-B4 classifier (**96.8% accuracy, AUC: 97.1%** across 38 disease classes) feeds into a **Qwen2.5-3B QLoRA-finetuned LLM** generating structured treatment plans with organic/chemical options, yield impact, and region-season advisory (**BERTScore F1: 81.68%**).
  - Published app on **Render** and **HuggingFace Spaces**; models released to HuggingFace Hub with W&B experiment tracking and a **pytest** suite covering API, model forward pass, and schema validation.
  - **Technologies**: PyTorch, EfficientNet-B4, Qwen2.5-3B, QLoRA (PEFT/TRL), FastAPI, Vite, HuggingFace Hub, W&B, Render

- **Multimodal Price Estimator** *Github*

Deep Learning Project Oct 2025 – Oct 2025

  - Developed an ensemble pricing model using **parallel MLPs** on **CoCa** and **SigLIP** multimodal embeddings, achieving an **sMAPE of 40.06%** on product price prediction.
  - Optimized the architecture for fast inference at **0.12s/sample** through efficient model design.
  - **Technologies**: PyTorch, CoCa, SigLIP, CLIP, Ensemble Learning

Technical Skills

- **Languages & Core:** Python, SQL, Data Structures & Algorithms, OOP
- **ML, DL & GenAI:** Machine Learning, Deep Learning, NLP, Transformers, LLMs, SLMs, VLMs, PyTorch, Scikit-learn, Pandas, NumPy
- **LLM Engineering:** Pretraining, SFT, RLHF, DPO, PEFT, LoRA, QLoRA, Quantization, Hugging Face (Transformers, PEFT, TRL)
- **RAG & Retrieval:** RAG, Hybrid Retrieval, Embeddings, Reranking, Vector Databases, GraphRAG, Agentic RAG
- **Agentic AI:** ReAct, Tool Calling, Multi-Agent Systems, Context Engineering, Agent Memory, AgentOps, LangChain, LangGraph, MCP, Human-in-the-Loop
- **AI Safety, Evaluation & Observability:** LLM Guardrails, Prompt Injection Defense, RAGAS, DeepEval, LangSmith, Observability, Tracing
- **MLOps, Deployment & Infrastructure:** FastAPI, Docker, MLflow, Weights & Biases, vLLM, AWS, PostgreSQL, Git

Certifications

- **Python Data Structures and Algorithms** – Udemy *2025*
- **Machine Learning and Data Science** – Udemy *2023*

Achievements

- **Secured AIR 121**−in GATE Data Science and Artificial Intelligence 2025 with GATE Score of 780. *2025*
- **Secured Rank 11**−Amazon ML Challenge 2025 among 82,000+ teams across the country. *2025*